

BARKING SANDS, HI, USAWMO: **911620**Lat: **22.033N**Lon: **159.783W**Elev: **23**StdP: **14.68**Time Zone: **-10.00 (W10)**Period: **96-19**WBAN: **22501****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	62.3	63.6	46.0	45.9	70.7	49.8	53.0	69.3	27.0	74.0	25.2	74.3	4.4	70	0.295

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.8	88.2	74.4	87.2	73.9	86.4	73.5	77.8	84.1	77.0	83.9	76.2	83.4	10.8	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.0	135.7	81.3	74.7	130.2	81.4	73.7	125.4	81.3	41.2	84.6	40.2	83.8	39.5	83.2	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.4	19.1	15.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	76.2	72.2	72.4	72.8	74.6	76.3	77.9	79.4	79.9	79.9	78.7	76.3	74.0
	DBStd	3.44	2.19	2.34	2.59	2.20	2.30	1.70	1.48	1.41	1.57	1.55	1.89	2.06
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	9571	690	628	706	738	814	837	911	928	896	891	788	744
	CDD65	4096	225	208	241	288	349	387	446	463	446	426	338	279
	CDH74	31530	978	902	1214	1756	2634	3330	4297	4613	4292	3687	2346	1481
	CDH80	7428	29	39	96	208	499	828	1299	1491	1388	1018	432	101
Wind	WSAvg	6.1	7.3	8.3	7.3	6.1	5.9	5.3	5.3	5.2	5.4	5.3	5.9	6.3
Precipitation	PrecAvg	59.1	6.5	5.2	6.7	4.4	3.8	3.0	3.3	3.7	3.5	5.0	6.8	7.6
	PrecMax	83.3	16.5	9.3	30.9	9.6	9.2	6.1	4.8	6.3	7.6	8.6	16.7	18.9
	PrecMin	37.4	1.5	1.0	1.8	2.2	1.5	1.1	2.4	1.9	2.1	2.0	3.1	1.5
	PrecStd	12.1	3.7	2.4	6.0	1.9	2.0	1.2	0.6	1.1	1.6	2.0	3.7	4.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.2	83.9	84.7	84.8	85.9	87.3	88.7	89.2	89.7	88.4	86.2	83.5
		MCWB	70.7	68.2	68.4	70.2	72.3	71.9	73.6	75.5	75.7	73.2	72.3	70.6
	2%	DB	80.9	81.4	82.2	83.3	84.8	86.0	87.2	87.8	87.9	86.8	84.5	82.2
		MCWB	70.2	69.4	68.5	70.3	71.7	71.7	73.6	74.7	75.1	73.2	71.8	70.7
	5%	DB	79.9	80.2	80.9	82.3	83.9	85.1	86.3	86.8	86.8	85.7	83.5	81.2
		MCWB	69.7	69.0	68.4	69.9	71.6	71.6	73.4	73.9	74.6	73.3	71.4	70.8
	10%	DB	78.9	78.9	79.7	81.2	82.9	84.1	85.4	85.8	85.9	84.6	82.6	80.3
		MCWB	69.1	68.4	68.2	69.5	71.3	71.6	73.2	73.2	74.0	73.1	71.0	70.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.6	74.8	73.7	75.4	77.4	75.4	77.8	78.4	79.3	78.2	77.1	76.8
		MCDB	78.7	78.7	79.6	79.3	82.1	81.4	84.5	85.7	84.3	83.1	81.5	79.6
	2%	WB	72.8	72.9	71.9	73.3	75.6	74.5	76.6	77.2	77.9	76.8	75.5	74.2
		MCDB	78.0	77.6	77.5	79.1	81.7	81.5	84.3	85.2	84.3	83.2	81.6	79.3
	5%	WB	71.5	71.5	70.8	72.1	73.8	73.5	75.7	76.3	76.9	75.8	74.2	73.0
		MCDB	77.4	76.9	76.9	78.5	80.7	81.7	83.7	84.3	84.0	82.8	80.9	78.9
	10%	WB	70.6	70.4	69.8	71.1	72.6	72.9	74.8	75.3	75.9	74.8	73.0	71.9
		MCDB	76.9	76.6	76.8	78.2	80.2	81.5	83.2	83.5	83.6	82.4	80.3	78.2
Mean Daily Temperature Range	5% DB	MDBR	13.3	13.0	13.3	13.2	13.0	13.2	13.1	12.8	13.0	12.9	13.3	12.8
		MCDBR	13.8	14.1	14.7	14.1	13.7	13.9	13.7	13.5	13.7	13.7	13.9	13.6
		MCWBR	8.5	8.7	8.2	7.8	8.0	7.6	7.9	8.0	8.1	8.3	8.6	8.7
	5% WB	MCDBR	12.6	11.7	12.4	12.2	12.6	12.9	12.6	12.3	12.1	12.5	12.6	11.7
		MCWBR	8.9	8.3	7.6	7.7	8.0	7.8	7.6	7.5	7.3	8.6	9.2	8.9
Clear-Sky Solar Irradiance	taub	0.354	0.357	0.363	0.396	0.387	0.375	0.375	0.373	0.366	0.369	0.361	0.349	
	taud	2.603	2.588	2.577	2.469	2.523	2.565	2.573	2.578	2.603	2.610	2.626	2.646	
	Ebn at Noon	292	297	298	288	287	288	288	291	293	289	287	290	
	Edh at Noon	28	30	32	36	34	32	32	32	31	29	28	26	
All-Sky Solar Radiation	RadAvg	1402	1565	1815	2070	2220	2316	2286	2199	2044	1693	1423	1284	
	RadStd	57	133	190	120	128	60	61	84	60	68	49	94	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A